

L3: Degree Distribution Moments

Average Degree

$$\bar{k} = \sum_{k=0}^{\text{maxdegree}} k p_k$$

First Moment

Second Moment

$$\overline{k^2} = \sum_{k=0}^{\text{maxdegree}} k^2 p_k$$

Variance

$$\sigma_k^2 = \overline{k^2} - (\bar{k})^2$$

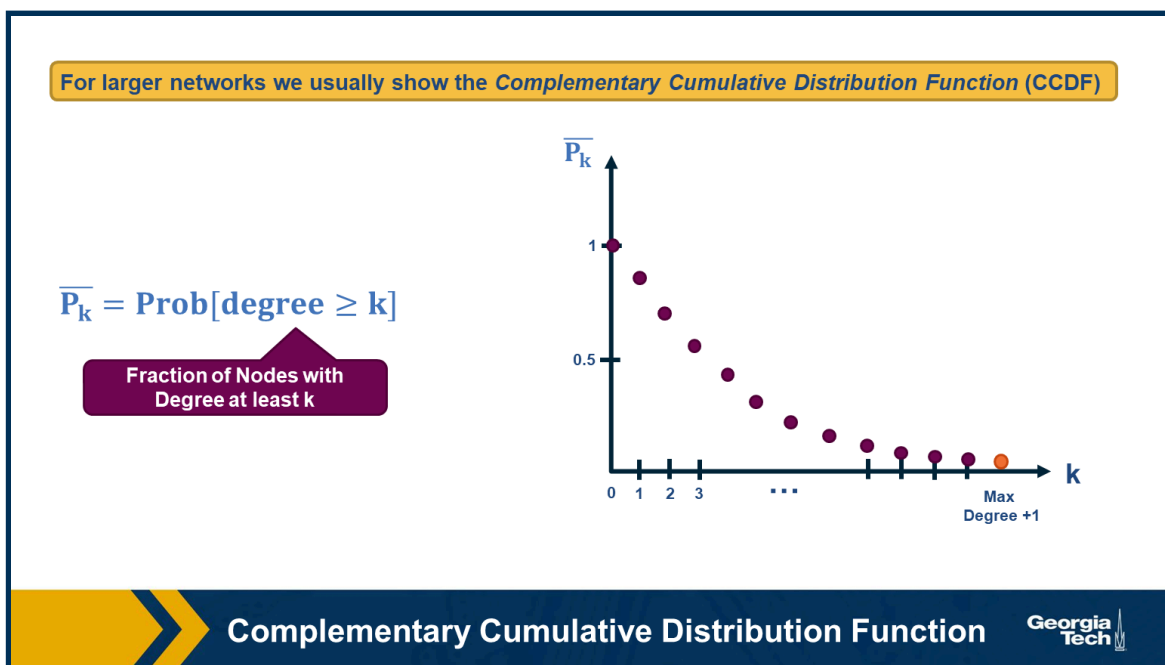
Degree Distribution Moments

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Recall that, given the probability distribution of a random variable, we can compute the first moment (*mean*), second moment, variance, etc.

The above formulas show the moments that we will mostly use in this course: the average degree, the second moment of the degree (*the average of the squared degrees*), and the variance of the degree distribution.

Complementary Cumulative Distribution Function



For larger networks, we usually do not show the empirical probability density function p_k . Instead, we show the probability that the degree is at least k , for any $k \geq 0$.

This is referred to as the Complementary Cumulative Distribution Function (*denoted as C-CDF*). Note that $\bar{P}_k = \text{Prob}[\text{degree} \geq k]$ is the sum of all p_x values for $x \geq k$.